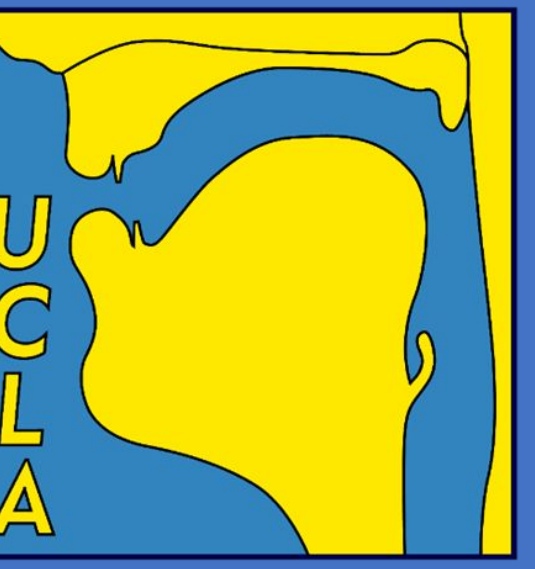




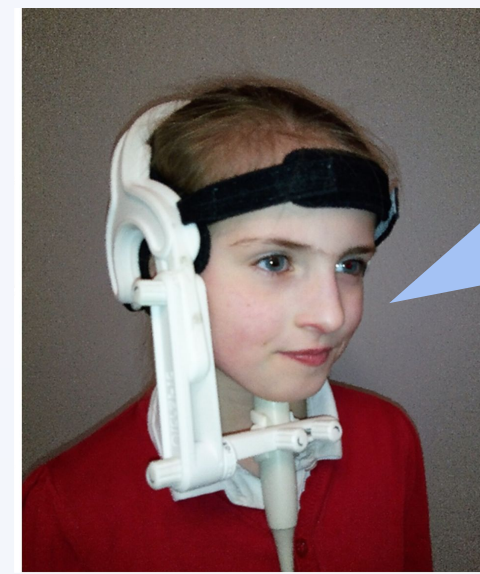
4pSC32: A systematic review and meta-analysis of the development of coarticulation in child speech



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BACKGROUND

- Coarticulation refers to the temporal and spatial overlap of speech gestures
 - Reflects speech planning^{1,2} and linguistic structure³
- Coarticulation can be measured using a variety of methods:
 - Articulatory⁴, Acoustic^{5,6}, Temporal⁷
- The degree of coarticulation changes and develops throughout childhood^{8,9}



(Cleand, 2023)

Methods of measuring coarticulation abound but results are still inconclusive!

Methodological differences across studies make it challenging to characterize how the degree of coarticulation develops throughout childhood.

RESEARCH QUESTIONS

1. How does the degree of coarticulation develop in children's speech between ages 3;0 – 10;11?

- The literature is mixed,^{4, 6, 10, 11} but recent reports suggest that **coarticulation decreases** as children develop¹²

2. Does the relationship between coarticulation and age vary by measurement technique?

- We hope not! But we will compare acoustic, articulatory, and temporal measurements.

SEARCH METHOD: CRITERIA FOR INCLUSION

- Peer-reviewed papers, studies, and proceedings (1984 – 2024)
- Typically developing monolingual children (3;0–10;11) acquiring any language
- Anticipatory and perseverative coarticulation in **CV** and **VC** sequences (i.e., not long-distance V.CV)
- Acoustic, articulatory, and temporal measures

DATA COLLECTED

- Type of study (13 acoustic; 9 articulatory; 1 temporal)^(Appendix III)
- Target unit (e.g., CV, VC)
- Target languages (e.g., English, Scottish English, German)
- Population information (number of subjects, gender, etc.)

STATISTICAL METHODS

- Normalized on 0.00001-1 scale, grouped by broad measure type and by unit/scale measure
- Binned participants into 2-year age bins
- Effect size (Hedges' g) estimated using meta-regression model
- Missing measures of variance and means imputed¹³:
 - Mean taken as intercept if no measures of variance reported

RESULTS

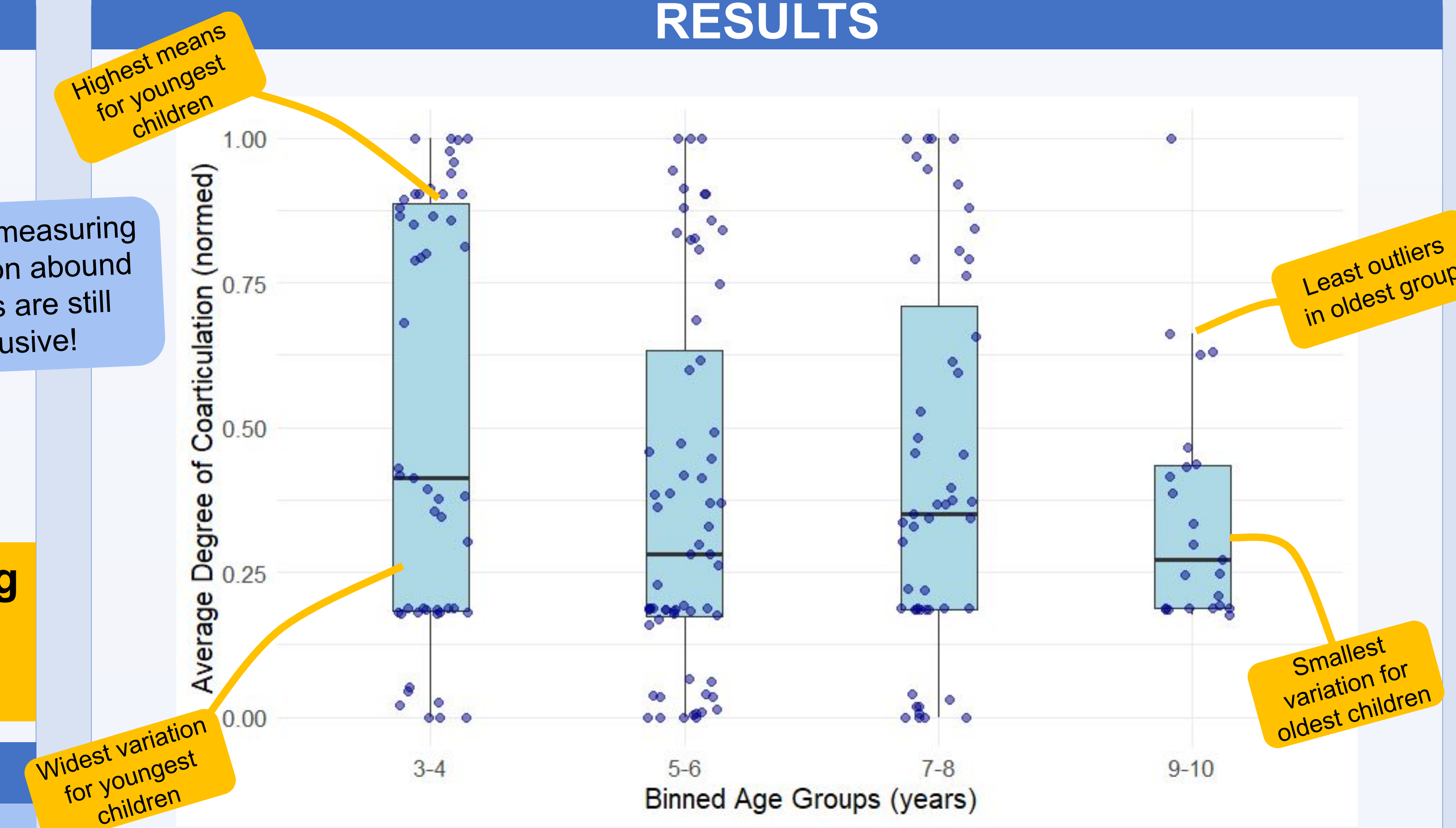


FIG 1: Distribution of Normalized Mean Measure (x) by Age Bin (y)

1. Visualisation of Age Effect:

- Normalized mean measure plotted for each age bin (FIG 1)
 - Variation in means decreases inversely with age group

2. Meta-Regression Model:

- Small but significant negative relationship between age and extent of coarticulation: children coarticulate less as they age
 - $\beta = -0.001$, $p < 0.001$
- Test for heterogeneity suggests high degree of bias between studies beyond sampling error
 - Broad measurement type (i.e., articulatory, temporal, acoustic) significantly moderates relationship ($p < 0.005$ for all); so age effects moderated by measurement method
 - Decreases systematically for acoustic data, but less inversely decreasing and lower means for articulatory

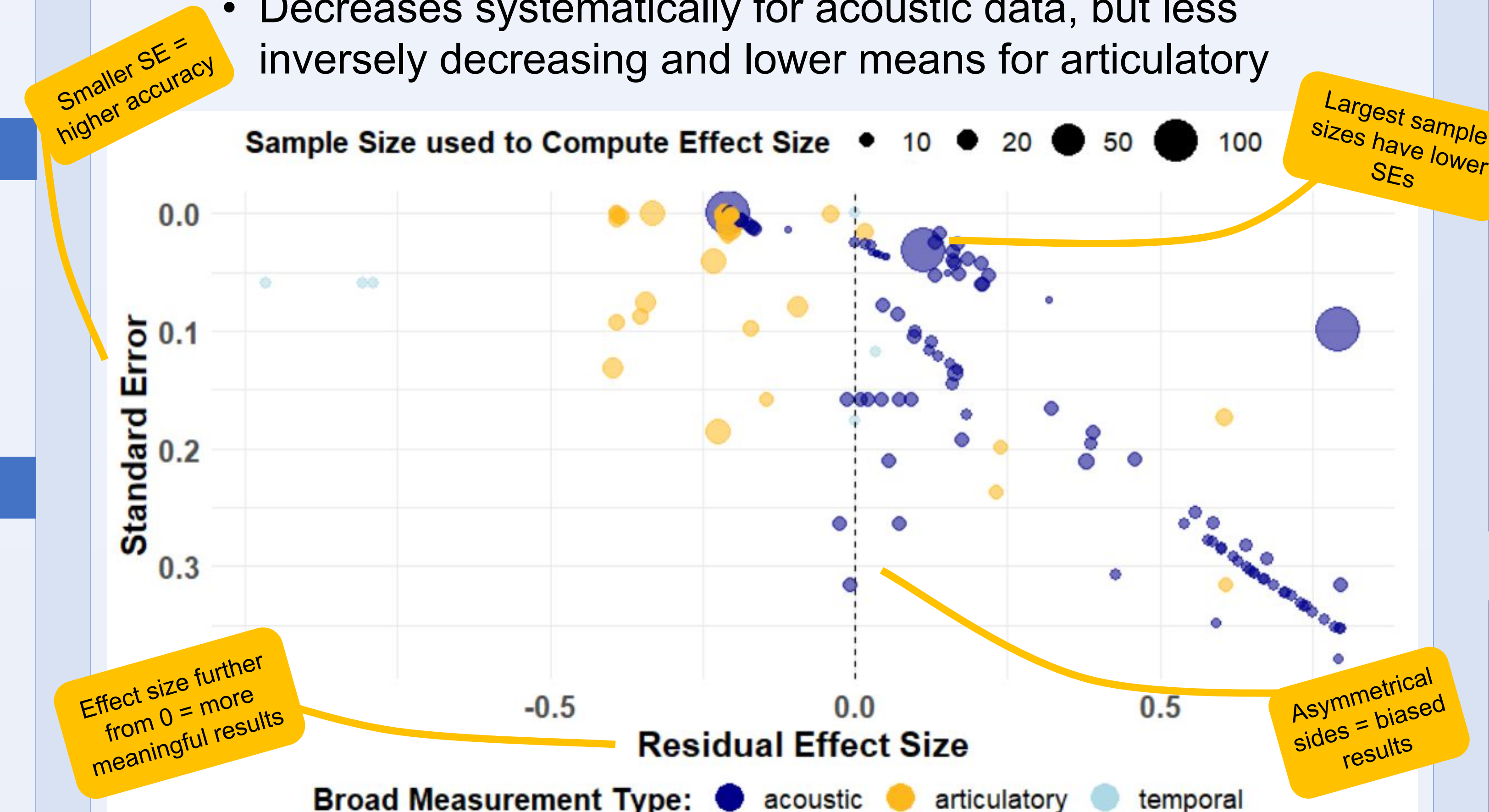


FIG 2: Relationship between Effect Size (x) and Standard Error (y)

3. Bias Derived from Sample Sizing

- Funnel plot (FIG 2) shows relationship between sample size, residual effect size, and measures of variance
 - Greater accuracy (smaller SE) with larger sample sizes
 - Asymmetry (dots not dispersed roughly equally across sides) demonstrates bias – study **over-estimates** effect
 - Significant colour differentiation across sides shows effect of broad measurement type
- Egger's test to measure funnel symmetry
 - Intercept = -0.23 ($p < .001$), indicates results biased by sample size- significant deviation from 0

Finding #1: Degree of coarticulation *decreases* between 3;0 – 10;11 years

- Coarticulatory patterns systematically **decrease** throughout childhood

Finding #2: Measurement type influences effect of coarticulation measures

- Broad measurement type (acoustic, articulatory) significantly moderates measures of variance when reporting coarticulation- clearest pattern for acoustic studies

Finding #3: Evidence of systematic publication bias

- Underpowered studies **report inflated effect sizes** and are **overrepresented** in the literature
- Robustness of results may be unreliable in studies that use small sample sizes

IMPLICATIONS AND FUTURE RESEARCH

- First quantitative synthesis of developmental coarticulation research
- Shows underpowered studies are less reliable, especially when considered in isolation
- Overall, evidence that coarticulation decreases over development and future work should prioritize higher-powered studies given evidence of publication bias

REFERENCES AND APPENDICES

